

2021 JC2 PRELIMINARY EXAMINATION

CANDIDATE
NAME

MARK SCHEME

CLASS

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INDEX
NUMBER

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H2 BIOLOGY

9744/02

Paper 2 Structured Questions

15 September 2021

Candidates answer on Question Paper.
No Additional Materials are required.

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

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/100	

This document consists of **27** printed pages and **1** blank page.

Answer **all** questions.

- 1 In 1934, two biologists Davson and Danielli published their suggestion for the structure of the cell surface membrane, as shown in Fig. 1.1.

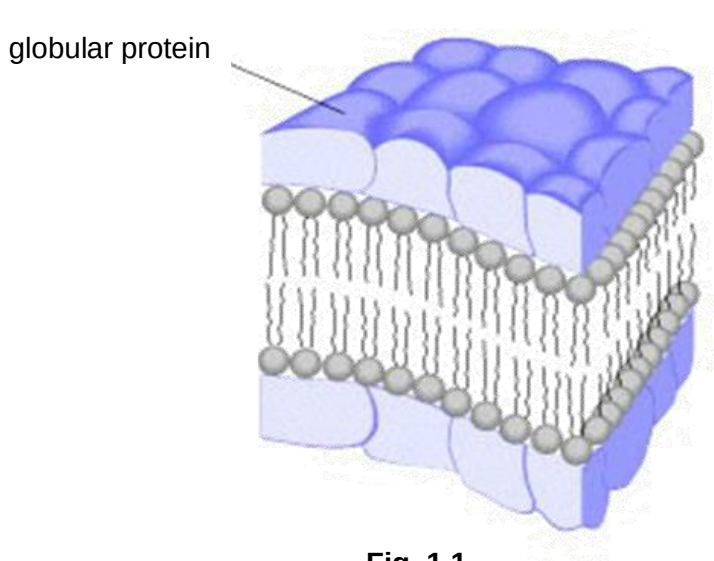


Fig. 1.1

- (a) (i) State one way in which the Davson-Danielli structure is similar to the fluid mosaic model and one way in which it differs from the fluid mosaic model.

[Similarity – max 1]

- Phospholipids are arranged in a bilayer
- AVP: Both contain peripheral proteins on the membrane

[Difference – max 1]

- The proteins in the Davson-Danielli structure are peripheral / outside the bilayer while the proteins in the fluid mosaic model are both integral and peripheral to the bilayer
- Proteins in the Davson-Danielli structure are in **uniform locations** across the surface of the membrane but proteins in the fluid mosaic model are found in **different locations and numbers** across the membrane
- In the fluid mosaic model, the membrane contains other molecules like cholesterol but the membrane in the Davson-Danielli structure does not.
- In the fluid mosaic model, the membrane contains glycoproteins / glycolipids but the membrane in the Davson-Danielli structure does not.

[2]

- (ii) Suggest two problems that the Davson-Danielli structure of the membrane would pose to the functioning of the cell.
- Limited regulation of membrane fluidity in the absence of cholesterol
 - Not able to transport hydrophilic/polar/charge substances across the membrane as there are no transmembrane proteins to serve as transport proteins.
 - Layer of globular proteins blocking simple diffusion of hydrophobic substances.
 - No transmembrane receptors to transmit signals from the extracellular environment
 - No glycoproteins for cell-to-cell recognition / adhesion

[2]

Many proteins such as the receptor tyrosine kinases are embedded in the cell surface membrane. Phosphatidylinositol 3-kinases (PI3K) signalling is initiated by the activation of a number of receptor tyrosine kinases such as the epidermal growth factor receptor (EGFR). Current estimates suggest that this signalling pathway is upregulated in 30–50% of prostate cancers.

Fig. 1.2 shows the PI3K signalling pathway.

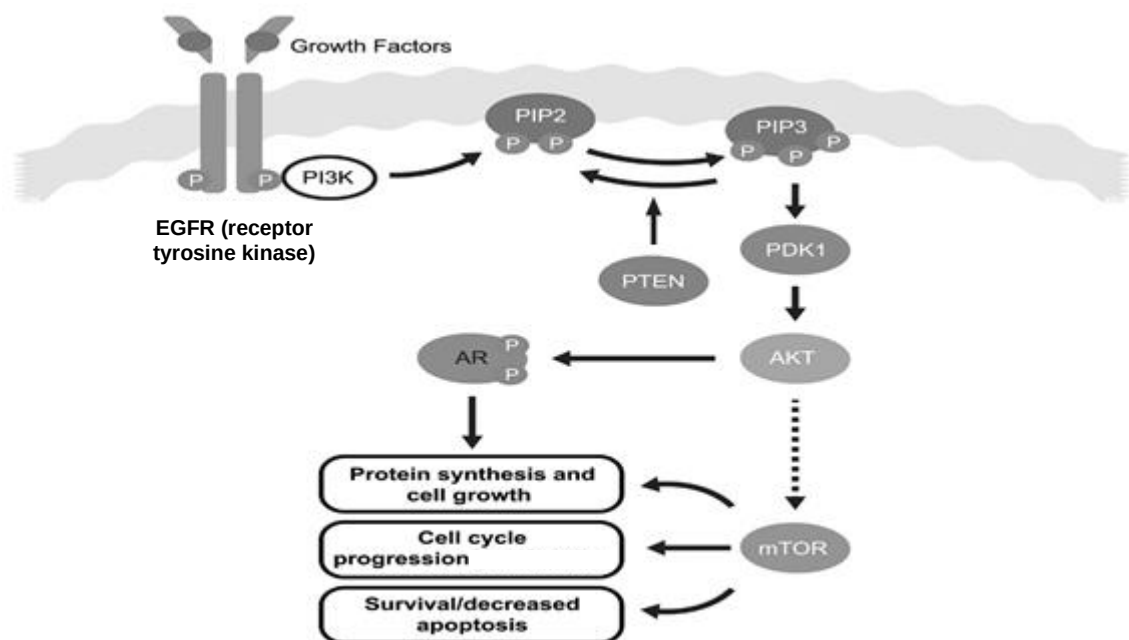


Fig. 1.2

(b) Based on biological knowledge and Fig. 1.2, describe what happens after binding of growth factors to EGFR to result in formation of a tumour mass.

- EGFR/ RTK **dimerize** and **cross phosphorylation** of tyrosine kinase residues on the intracellular tail occurs;
- PI3K is **activated (via phosphorylation)** by tyrosine kinase residues and goes on to activate PIP2;
- PIP2 is **phosphorylated** to PIP3 + PIP3 activates PDK1 which in turn, activates AKT;
- Signal amplification/ Each PDK1 can phosphorylate more than one AKT, resulting in signal amplification;
- Uncontrolled cell division, rate of cell division exceeds cell death.

[4]

(c) An upregulation of the PI3K signalling pathway is mostly due to the deletion of the tumour suppressor gene, *PTEN*.

In order for the PI3K signaling pathway to be upregulated, both copies of the *PTEN* gene needs to be deleted.

Explain this observation.

Loss-of-function mutation is a recessive mutation/ One copy of functioning *PTEN* gene will be able to produce sufficient quantities of functional *PTEN*; [1]

(d) A 40 year old man decides to do a genetic test. On his report, it was stated that only a homozygous deletion of *PTEN* gene has been detected in the sample of cheek cells that he sent. He concluded that he has oral cancer.

Comment on the validity of his conclusion.

- not valid, cancer is a **multi-step process** where there must be **accumulated mutations** in a single cell such as (at least 1 example) gain-of-function mutation of proto-oncogene/ activation of **telomerase** gene/ activation of genes involved in **angiogenesis**; [1]

[Total: 10]

- 2 Collagen is the most abundant protein found in the human body, and is majorly found in connective tissue.

(a) Glycine is a common amino acid found in collagen.

Describe the role of glycine in a collagen molecule. [2]

[any 2]

1. **Small size/ small R group/ hydrogen atom as R group** places glycine at the **centre of the triple helix**
2. Allowing for **tighter packing** of the three polypeptide chains
3. **-NH group** of many **glycines** on one chain forms **inter-chain hydrogen bonds** with the **-C=O groups** of peptide bonds on one of the other two polypeptide chains.

Collagen “unwinds” or “shrinks” under heat treatment.

Fig. 2.1 shows how the appearance of a collagen molecule changes after heat treatment, while Fig. 2.2 shows experimental data of how its structure changes with temperature.



before heat treatment – normal



after heat treatment – contracted

Fig. 2.1

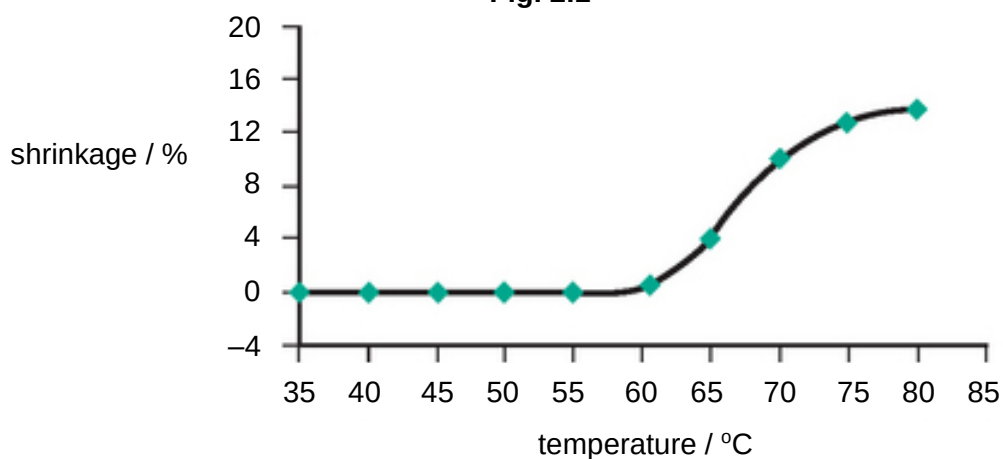


Fig. 2.2

- (b) With reference to the information including Fig. 2.1 and Fig. 2.2,
- (i) state why collagen is classified as a quaternary protein. [1]
 1. It comprises three polypeptide chains held together by inter-chain hydrogen bonds.
 - (ii) describe **and** explain how the structure of a collagen molecule is affected when temperature increases from 35°C to 80°C. [4]
 1. **From 35 - 60 °C, no shrinkage occurs/** shrinkage remains constant at **0%**;
 2. as tropocollagen/ triple helix structure remains intact due to extensive/ numerous inter-chain hydrogen bonds.
 3. As temperature increases from **60 °C to 80 °C**, shrinkage **increases** from **0 to 14%**;
 4. as **weak interchain hydrogen bonds** are **disrupted/ broken** by heat.
 - (c) State how a **named** property of collagen is affected with increasing temperature. [1]
 1. Tensile strength is **reduced**
 - (d) Suggest why the collagen molecule in Fig. 2.1 does not unwind completely even at high temperatures beyond 80°C. [1]

[any 1]

 1. Collagen chains are **long**, thus **still easily tangled** with one another/ OWTTE (other words to that effect).
 2. **Strong covalent cross-links** still present at ends within tropocollagen (not disrupted by heat). (Ref: <https://www.sciencedirect.com/topics/nursing-and-health-professions/tropocollagen>)
 3. **AVP**

[Total: 9]

- 3 In 2008, a woman had her damaged left bronchus replaced by one constructed using her own stem cells. A section of the trachea from a donor was treated in order to remove the cells of the donor, leaving only a framework of collagen. Adult stem cells derived from the woman's own bone marrow were then added to this collagen framework. Samples of these stem cells, together with cells derived from the lining of her trachea, were placed in a bioreactor for four days during which they multiplied to coat the collagen framework. This was used to replace the damaged section of her bronchus. A month later the transplanted tissue had developed its own blood supply.

This was claimed to be the first successful transplant using tissues derived from the patient's own stem cells.

- (a) Explain the advantages of using a patient's own stem cells to treat a damaged bronchus, as described in the passage, rather than transplanting a complete bronchus. [3]

Any 3 pts

1. **No immune/ tissue rejection**/patient's own cells will not be **recognised as foreign** by the body's immune system
2. There is no need for **immunosuppressants / immunosuppressive** drugs
3. Using patient's stem cells **overcomes the need of finding suitable donors**/remove the need to do tissue / blood match
4. **Bronchus cells can be obtained/cultured in large quantities** when using patient's own stem cells
5. Healthy culture of patient's stem cells can be **stored for future needs**

R: arguments on cost (lack of information provided to determine which procedure is more costly)

- (b) Stem cells from the human bone marrow that are involved in blood cell formation are described as multipotent, rather than totipotent.

Compare multipotent and totipotent stem cells. [2]

Similarity – any 1

1. Capable of differentiating to give rise to specialised cells
2. Are unspecialised
3. Capable of self-renewal via mitosis

Difference

	Totipotent SCs	Multipotent SCs
1.	able to differentiate into / give rise to <u>all cell types to form a whole organism including extraembryonic tissues</u>	able to differentiate into / give rise to a <u>limited range of related cell types</u>
2.	found in zygote	found in developed tissues or organs

- (c) Some fully differentiated cells can be stimulated to change back into stem cells in tissue culture. Such cells are called induced pluripotent stem cells (iPS cells).

In experiments with mice, it was discovered that the introduction of four genes would cause certain fully differentiated cells to change to iPS cells. There is evidence to suggest that the introduction of the four genes caused an increase in the production of telomerase reverse transcriptase (TERT) in the fully differentiated cells.

Explain how TERT may help to change the cells back into stem cells. [3]

1. TERT plays a part in **lengthening / extending** the length of **telomeres**
2. **Overcomes** (**not prevent**) **end replication problem** or critical/essential genes at the **end** of chromosome are not eroded/lost
3. Cells do not undergo **apoptosis** or **senescence**
4. Ensuring that the iPS cells have the capacity to **divide continuously/ indefinitely/for long periods of time** or removes the natural limit number of times a chromosome can replicate / number of times cells can divide

- (d) Suggest a potential risk with such treatments involving production of TERT in cells. [1]

1. **Ref to uncontrolled cell division/ growth of tumour/ risk of cancer**

[Total: 9]

- 4 *Escherichia coli* (*E. coli*) is a species of bacteria that is commonly found in the gut of humans and warm-blooded animals. While most strains of *E. coli* are harmless, some strains such as Shiga toxin-producing *E. coli* (STEC), can cause severe and potentially fatal foodborne disease. STEC is transmitted to humans primarily through consumption of contaminated foods, such as raw or undercooked ground meat products.

Fig. 4.1 is a micrograph of *E. coli* cells.

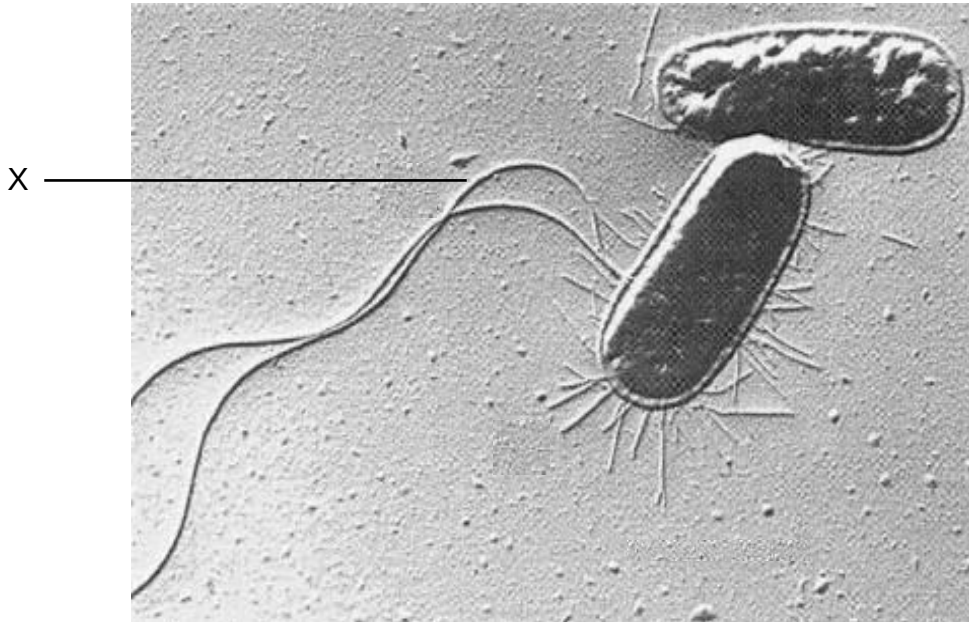


Fig. 4.1

- (a) Identify structure X. [1]
flagellum
- (b) During its life cycle, Shiga toxin-producing *E. coli* (STEC) is exposed to a variety of acidic conditions. Survival in the acidic environments within the hosts is characterised by the induction of several *E. coli* genes upon exposure to acidic conditions.

Fig. 4.2 shows an operon found in STEC involved in the regulation of pH.

1. CadC is a transmembrane protein that is encoded by the *cadC* gene. In the presence of lysine under acidic conditions, CadC is activated.
2. Expression of *cadB* and *cadA* genes is induced.
3. CadA and CadB proteins are produced to help regulate intracellular and extracellular pH. CadA is an enzyme and CadB is a transmembrane protein.
4. Lysine is converted to cadaverine which is a proton acceptor.

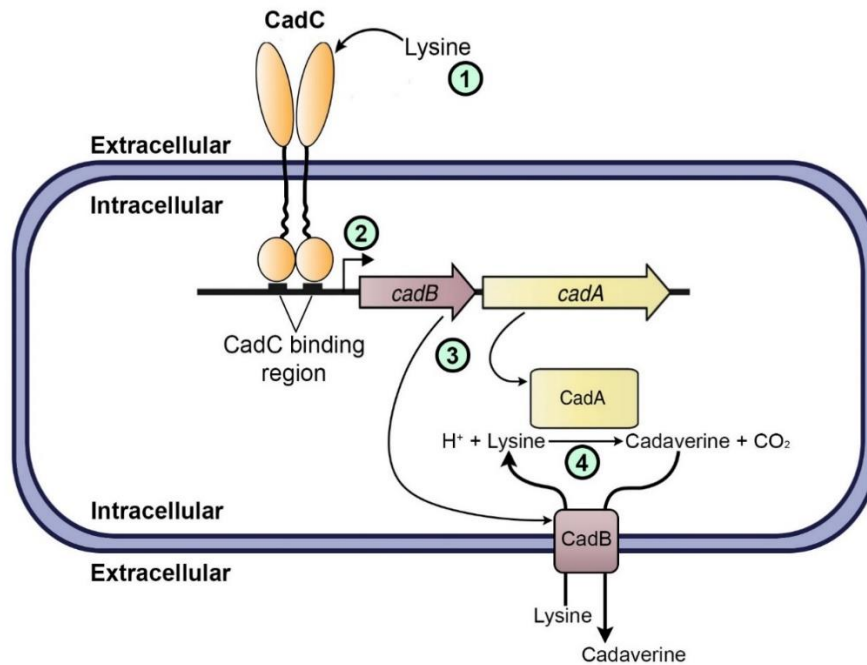


Fig. 4.2

(i) The operon in Fig. 4.2 is expressed in an inducible manner. Explain the meaning of 'inducible'. [1]

1. Inducible means **structural genes** of the operon is **not expressed** by default/ can be **stimulated to be expressed** in the presence of an **inducer** molecule

(ii) With reference to Fig. 4.2, explain how an activated CadC protein leads to the regulation of extracellular pH in STEC when exposed to acidic conditions. [4]

Any 4

1. Activated CadC has a conformation/DNA binding site that is **complementary in shape** to CadC binding region on the DNA
2. Activated CadC acts as an activator that leads to positive regulation/increased rate of transcription/expression of *cadB* and *cadA* structural genes
3. **CadA** protein **converts lysine to cadaverine**
4. **CadB** protein act as a **channel/carrier protein** to **transport cadaverine** to **extracellular** region
5. Cadaverine will then **accept protons** thus increasing pH/decreasing H⁺ concentration in the extracellular environment

- (c) STEC produces toxins known as Shiga-toxins which prevent protein synthesis in the host cell and results in cell death. When infected, patients require both anti-toxin and antibiotic treatment.

Suggest why a combination of anti-toxin and antibiotic treatment can more effectively treat STEC-infected patients. [3]

1. Anti-toxin ensures **survival/restores normal functioning** of host cells **while** antibiotic **kills/ prevents growth** of **STEC** cells [idea of targeting 2 different aspects]
2. Anti-toxin therapy **degrades/removes/neutralises toxins** so that it is unable to inhibit an enzyme involved in transcription or translation [Mechanism of anti-toxin therapy]
3. Antibiotic therapy **inhibit cell wall synthesis/nucleic acid synthesis/protein synthesis** to prevent growth of/kills STEC cells [Mechanism of antibiotic therapy]

[Total: 9]

- 5 The Ebola outbreak in West Africa is the world's deadliest to date and the World Health Organization (WHO) declared an international health emergency in response. According to the UN, 1013 people have died in Guinea, Liberia, Sierra Leone and Nigeria in 2014.

Fig. 5.1 below shows some of the main structural features of the Ebola virus.

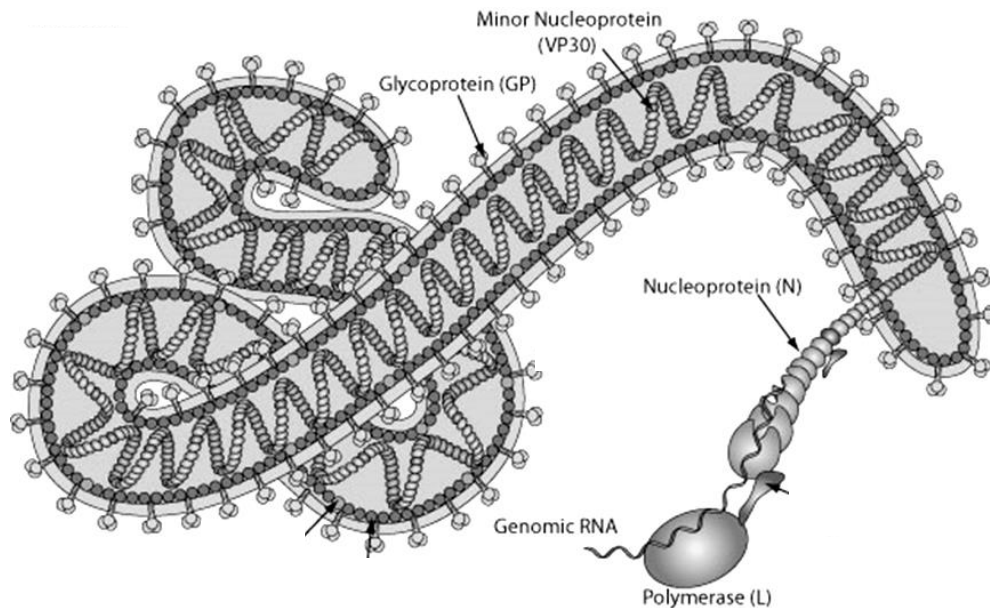


Fig. 5.1

- (a) The Ebola virus (EBOV) is similar to the influenza virus in many ways. For example, EBOV infects the host cell in the same way as influenza virus does.

Using biological knowledge of influenza virus, suggest how EBOV may infect a host cell.

- GP binds to **specific receptor** molecules on **host cell surface membrane**;
- virus enters via **vesicle/ endosome** through the invagination of the cell surface membrane by the process **endocytosis**;
- Viral envelope fuses with endosome's membrane, exposing the **capsid** to digestion by cellular enzyme and releasing the **genomic RNA**;

[3]

- (b) The genome of EBOV contains one molecule of linear, negative sense RNA which is 19kb long. The genome encodes seven proteins, which include nucleoprotein (N), VP30, glycoprotein (GP) and L protein. In particular, L protein is directly involved in genome replication.

Describe the role of L protein.

- L protein is an **RNA-dependent RNA polymerase**;
- **transcribes/ synthesises** negative sense RNA into **positive strand RNA**/ uses the negative sense RNA as a template to synthesise a positive strand;
- uses the **positive strand RNA** as a template for transcription of **more negative strand** RNA to serve as **genome** OR so that the positive sense RNA can be translated to virus's enzymatic and structural proteins;
- Host cell lack RNA dependent RNA polymerase;

[3]

- (c) Despite Ebola being a highly virulent disease in West Africa, suggest why there is little progress in drug development.
- Antigenic drift/ since RNA-dependent RNA polymerase has **no proof-reading in replication**, this makes development of vaccine difficult as **surface antigens change at a high rate**;
 - research is expensive, and drug has to be given cheaply because disease is in 3rd world country;
 - Not as lucrative because only some African countries are affected;
 - Disease progresses quickly, therefore lack of humans for clinical trials for drugs which alleviate symptoms;
 - lack of willingness for vaccine trials;

[3]

[Total: 9]

- 6 Mitochondrial DNA (mtDNA) is a double-stranded molecule that consists of approximately 33,000 nucleotides.

- (a) The two strands of mtDNA differ in their base composition, thus allowing both strands to be separated to give a heavy and a light strand **by** density centrifugation.

Suggest how the heavy strand differs from the light strand. [1]

1. The heavy strand contains more **purines/ guanine** (accept: adenine) than the light strand.

The replication of mtDNA involves the Twinkle protein, which has both helicase and primase activity.

In humans, the Twinkle protein is encoded by the *TWINK* gene that is located in the long arm of chromosome 10.

- (b) Fig. 6.1 is a simplified diagram of a human cell, which shows some key structures typically present.

On Fig. 6.1, use a labelling line and

- (i) the letter **P** to label the precise location where transcription of the *TWINK* gene occurs. [1]
- (ii) the letter **Q** to label the precise location where the Twinkle protein carries out its function. [1]

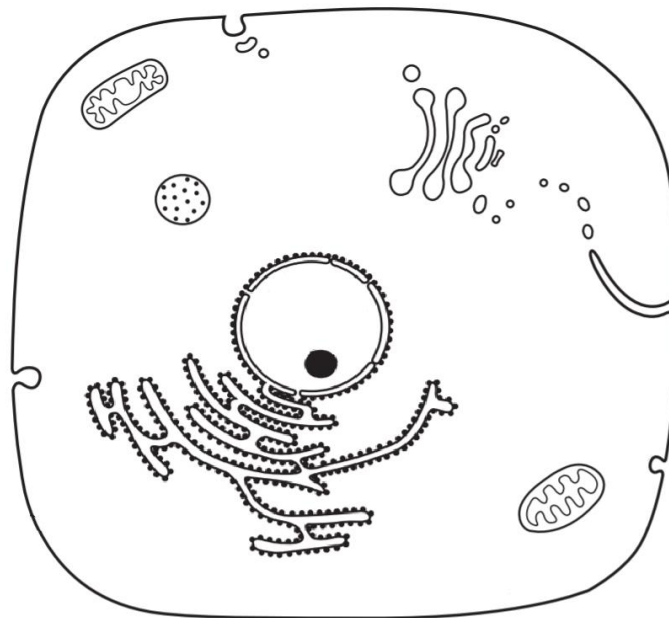


Fig. 6.1

1. labelling line with "P" indicating inside of the nucleus (Reject: labelling line touching nuclear envelope)
2. labelling line with "Q" indicating matrix of either mitochondria (Reject: labelling line touching outer mitochondrial membrane/ intermembrane space)

- (c) Explain the functions of the Twinkle protein during the replication of mtDNA. [3]

*[any 3]
helicase*

1. **separates/ unwinds and unzips/ breaks hydrogen bonds** within double-stranded mtDNA
2. to form **single-stranded** DNA strands which serve as **template**

primase

3. catalyses the formation of an **RNA primer**
4. which provides a **3' OH** end for **DNA polymerase**

- (d) The *TWINK* gene is also used to make the Twinky protein. While the function of this protein is unknown, some research suggest that it functions similarly to the Twinkle protein.

Explain how the single *TWINK* gene can be used to make two proteins. [2]

1. via **alternative splicing of pre-mRNA/ to form mature mRNA/ of mRNA post-transcription**
2. where **different splice sites are cut at/ only some splice sites are cut at** such that **different combinations of exons** are joined together/ result on mature mRNA.
(These mature mRNAs are then translated to form the two proteins)

- (e) Genes found on mtDNA are often examined to establish evolutionary relationships among organisms. Sequences from the same gene across multiple organisms are aligned to compare changes in this region of DNA.

Suggest why mitochondrial genes are often used to study changes in DNA sequences across organisms. [3]

1. mtDNA is **present in most organisms** since the mitochondrion is **essential** for aerobic **respiration**, hence serving as a good basis for comparison
2. Differences between **mitochondrial DNA** between two or more species are due **only to mutations**, as **recombination** (accept: crossing over/ meiosis) of **mtDNA does not occur**. *(Differences in mt gene sequences thus indicate only the mutations that have accumulated since divergence from a common ancestor.)*
3. Mitochondrial gene sequences tend to **change/ accumulate mutations very slowly**, hence useful for studying evolutionary changes (divergence) over a **longer time scale**
4. **Numerous mitochondria** are present in each cell, hence **fewer samples** for analysis required/ **easier to obtain mtDNA** for analysis

[Total: 11]

- 7 George Schull, a botanist at Princeton University, conducted a genetic study of a common weed known as shepherd's purse, *Capsella bursa-pastoris*. He studied the shape of its fruit, which could be heart-shaped or narrow respectively, as shown in Fig. 7.1.



Fig. 7.1

When he crossed a pure-breeding plant with heart-shaped fruit to a pure-breeding plant with narrow fruit, the F1 generation all had heart-shaped fruit. When the F1 generation was self-fertilised, the numbers of F2 generation with the respective shaped fruit were recorded as follows:

heart-shaped fruit	2251
narrow fruit	150

Fig. 7.2 shows the possible biochemical pathway that results in the observation made in the F2 generation.

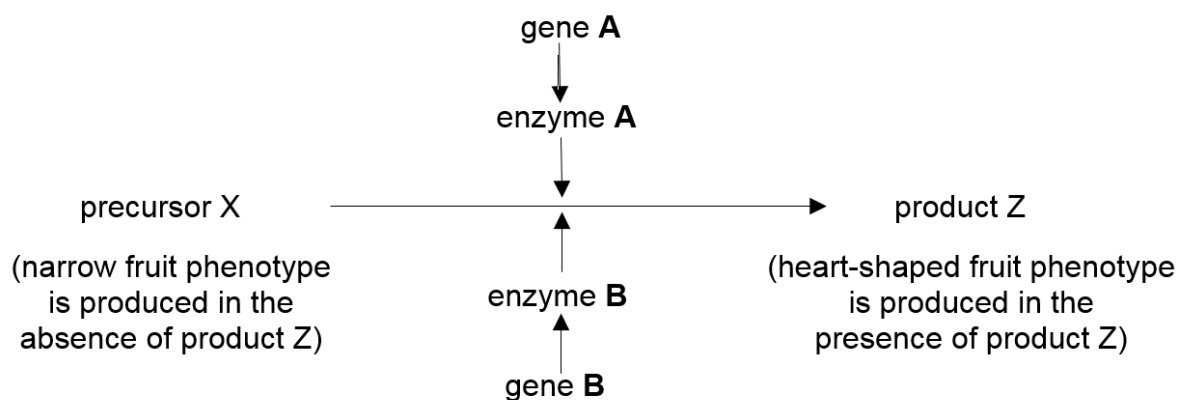


Fig. 7.2

(a) Explain the type of gene interaction observed in this context.

- The type of gene interaction is epistasis;
- Allele A is epistatic to alleles B and b, allele B is epistatic to alleles A and a;
- The presence of at least 1 dominant allele of either gene A or B causes the conversion of precursor X to product Z hence, allowing the formation of heart-shaped fruits;

OR

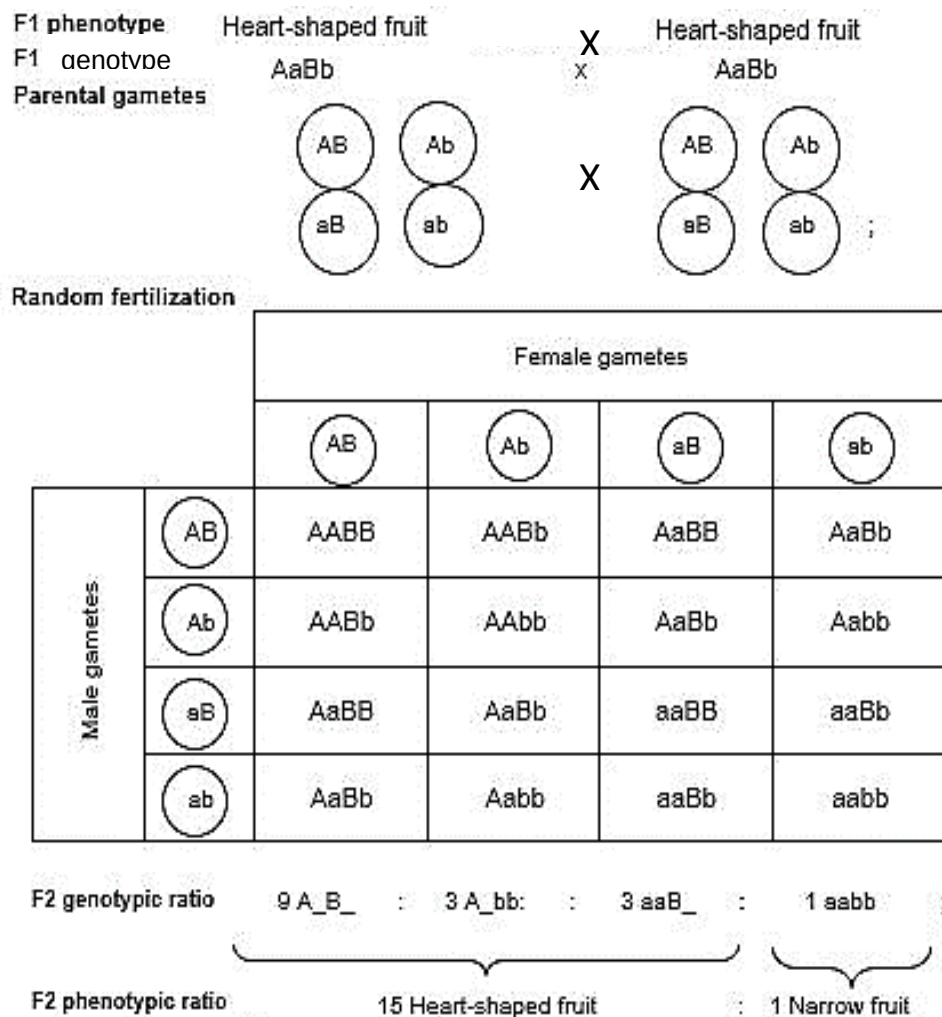
- If the genotypes at both loci are homozygous recessive, the conversion of precursor X to product Z does not occur;

[3]

(b) Draw a genetic diagram to explain the results of the F₂ generation in the space provided.

Include an appropriate legend using the symbols **A**, **a**, **B** and **b**.

Let A and B be the dominant alleles for the production of heart-shaped fruit
Let a and b be the recessive alleles for the production of narrow fruit;



[5]

- (c) Describe how *Capsella sp.* plants with heart-shaped fruits can be confirmed to be homozygous dominant in at least one gene locus.
- Perform a testcross using homozygous recessive plant/ narrow fruit/ aabb;;
 - If the plants with heart-shaped fruits are homozygous dominant in at least one gene locus, **all** offspring produced will only have heart-shaped fruits;;.

[2]

[Total: 10]

8 Yeast cells carry out respiration to produce ATP for cellular activities.

(a) In a yeast cell, molecules of glucose are phosphorylated at the start of glycolysis.

Suggest why the phosphorylated glucose molecules do not diffuse out of the cell. [3]
[any 3]

1. phosphorylated glucose is **too big** to diffuse through (transient pores in) cell surface **membrane/ channel** proteins that transport glucose
2. phosphorylated glucose is **polar/ hydrophilic** hence unable to diffuse across **hydrophobic** cell surface membrane
3. **no specific transport/ carrier/ channel protein** for phosphorylated glucose OR phosphorylated glucose is **not complementary** in shape hence unable to bind to **carrier** protein that transports glucose by facilitated diffusion
4. AVP, e.g. phosphorylated glucose is used up in glycolysis as soon as it is made

(b) Sometimes, yeast cells will need to carry out respiration in anaerobic conditions to produce ATP.

Explain why the respiration of glucose in anaerobic conditions produces less ATP than in aerobic conditions. [4]

[any 4]

in anaerobic conditions:

1. **substrate-level phosphorylation** during glycolysis is the only process which occurs that produces ATP
2. Ref. to net amount of **2 ATP** molecules **per glucose** produced during **glycolysis**
3. **pyruvate** is then converted to **ethanol** during alcoholic **fermentation**, where **no ATP is produced**
4. **oxygen** (molecule) is **not available/ present** as **final electron acceptor**
5. hence **chemiosmosis/ oxidative phosphorylation does not occur**, which would **produce most ATP/ 34 ATP molecules per glucose in aerobic conditions**
6. AVP, e.g. ref. to **ethanol** being **energy-rich**

(c) One of the by-products of respiration is hydrogen peroxide, which can have harmful effects in yeast cells at high concentrations. Hydrogen peroxide is rapidly broken down to water and oxygen by the enzyme catalase.

A student is asked to carry out an investigation into the effect of hydrogen peroxide concentration on the rate of its breakdown by catalase.

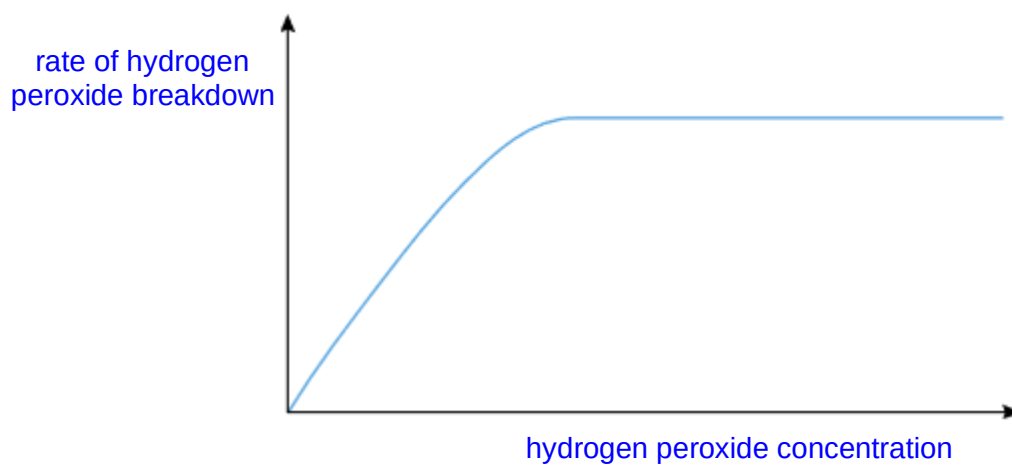
(i) Outline the key steps in this investigation including how the student can measure the rate of hydrogen peroxide breakdown. [3]

1. A **range** of hydrogen peroxide **concentrations** (or state example of range, e.g. 1.0%, 0.8%, 0.6%, 0.4%, 0.2%) is created by **simple/ serial dilution**
2. In different test tubes, add the **same/ fixed volume** (or specify volume, e.g. 2.0 cm³) of the prepared **concentrations of hydrogen peroxide**, before adding the **same/ fixed volume** of **catalase solution** (or specify volume, e.g. 1.0 cm³) (starting the stopwatch to time upon the addition of enzyme)
3. **Oxygen gas evolved** over a **fixed time period** (or specify time period, e.g. 5 min) can be collected in a **gas syringe**, and **volume** measured (Accept: number of bubbles evolved over fixed time period). This indicates the rate of hydrogen peroxide breakdown.

(ii) Predict the expected result by sketching a graph in the space below.

Include appropriate labels for the graph axes. [2]

- correct axes labels; y-axis: rate of hydrogen peroxide breakdown, x-axis: hydrogen peroxide concentration
- correct shape of graph



[Total: 12]

- 9 Fig. 9.1 shows the nucleotide base sequence of a length of DNA from the gene coding for a cytochrome from a modern human, *Homo sapiens*, and the corresponding base sequences from a Neanderthal and four other primate species.

Neanderthals were human or human-like animals that became extinct about 30 000 years ago and are known only from fossil remains. Chimpanzee, orangutan and gorilla are species of ape, while the macaque is a type of monkey.

modern human	ATGACCCCAATACGC AAAATTAAACCCCTAATAAAATTAATTAACCACTCA
Neanderthal	ATGACCCCAATACGC AAAATTAAACCCCTAATAAAATTAATTAACCACTCA
chimpanzee	ATGACCCCAACACGC AAAATTAAACCCCTAATAAAATTAATTAATCACTCA
orangutan	ATGACCTCAACACGTAAAATCAACCCCTAATAAAATTAATCAACCACTCA
gorilla	ATGACCCCTATACGC AAAACTAAACCCCTAACAAAATAATTAACCACTCA
macaque	ATGACTCCAATACGC AAATCCAACCCACTAACAAAATAATTAATCGCTCC

Fig. 9.1

- (a) Sometimes it is possible to obtain small quantities of DNA from fossils.

Name the technique that is used to provide multiple copies of such DNA molecules for analysis.

polymerase chain reaction

[1]

- (b) The scientific name for the Neanderthal is considered to be either *Homo sapiens* or *Homo neanderthalensis*.

A scientist considers that *Homo sapiens* is the correct name to use.

State the assumption that the scientist is making and explain why it is difficult to confirm this assumption.

Assumption:

- that Neanderthals were, a subspecies of / the same species as, *Homo sapiens* / modern humans;
- that Neanderthals and modern humans could interbreed to produce fertile offspring or any other species definition criteria;

Explanation:

- it is not possible to confirm that they were able to interbreed with *Homo sapiens* ;
- nor that if they interbred, any hybrids produced would be fertile ;
- it is not possible to obtain sufficient, morphological / physiological / behavioural / ecological / biochemical data ;
- only fragments of DNA are available / it is not possible to obtain sufficient DNA for complete analysis ;
- small sample of Neanderthals ;

[3]

- (c) Suggest what the data in Fig. 9.1 indicate about the evolutionary relationships between modern humans, Neanderthals and the other primate species.

- all the species share a recent common ancestor ;
- Humans and Neanderthals have the same sequence and are therefore more closely related/ diverged from a recent common ancestor;

Quote data:

1. the smaller the number of differences the, more closely related / more recently - **chimpanzees** are phylogenetically closest to Homo sapiens / humans /Neanderthals
2. **gorillas** and **orangutans** are more distantly related (to humans) than chimpanzees but nearer than macaques
3. **macaques**, are the most distantly related to humans (compared to any of the other species) / constitute the outgroup

	modern human	Neanderthal	chimpanzee	orangutan	gorilla	macaque
modern human		0	2	5	4	10
Neanderthal	0		2	5	4	10
chimpanzee	2	2		5	6	10
orangutan	5	5	5		9	13
gorilla	4	4	6	9		9
macaque	10	10	10	13	9	

[3]

- (d) Explain why any conclusions derived from the information in Fig. 9.1 need to be treated with caution.

Any 3

1. Idea that the **length of nucleotide sequence is only a small fraction of the cytochrome gene** (17 amino acids long), which is likely to be a larger protein
2. Only **one gene out of the entire genome** and is hence not representative of entire gene/genome
3. Idea that sequences shown come from **one individual of each species** and might not be representative of the population [small sample size]
4. There might be **slight variation between species**
5. AVP: possibility of **multiple substitutions at the same base** and therefore actual/true number of mutations not accurately reflected

[3]

[Total: 10]

- 10 (a) Complete the table below to show the disease to which the statements apply. Put a tick (✓) if the statement applies. [3]

statement	disease		
	small pox	sickle cell anaemia	tuberculosis
causative agent is a bacterium			✓
causative agent is a virus	✓		
is an infectious disease	✓		✓
is a genetic disease		✓	
is preventable by vaccines	✓		✓

Correct ticks for each disease – 1m for each correct set of ticks

- (b) In humans, three genes code for the production of antibodies:

- Heavy chain gene, which comprises 50 variable segments, 25 diversity segments and 6 joining segments.
- Kappa light chain gene, which comprises 40 variable segments and 5 joining segments.
- Lambda light chain gene, which comprises 30 variable segments and 4 joining segments.

A student calculated the total number of possible antibodies that vary at their antigen-binding site using the equation below:

$$(50 \times 25 \times 6) \times (40 \times 5) \times (30 \times 4) = 1.8 \times 10^8$$

- (i) Explain why the student's method of calculation is incorrect **and** show the correct answer with working. [4]

1. The student's incorrect method suggests that each antigen-binding site is made up of one H chain, one κ L chain and one λ L chain (or 2 different types of L chain)
2. The antigen-binding site is made up of one heavy chain and one light chain.
3. Only either the kappa or lambda light chain gene is expressed in each B cell

[Compulsory]

4. $[(50 \times 25 \times 6) \times (40 \times 5)] + [(50 \times 25 \times 6) \times (30 \times 4)] = \underline{2.4 \times 10^6}$

Total number of antibodies _____

- (ii) Suggest why the total number that you have calculated in (b)(i) is an underestimate of the actual number that may be produced in the human body. [2]

1. Somatic hyper-mutation occurs during B cell proliferation
2. Results in point mutations on the arranged VDJ gene segments of heavy chain gene or VJ gene segments of light chain gene that codes for variable domains of heavy and light chains to result in a different conformation of antigen binding site

Reject: Class switching

Fig. 10.1 shows a naïve B cell and a plasma cell. Plasma cells are derived from a naïve B cell.

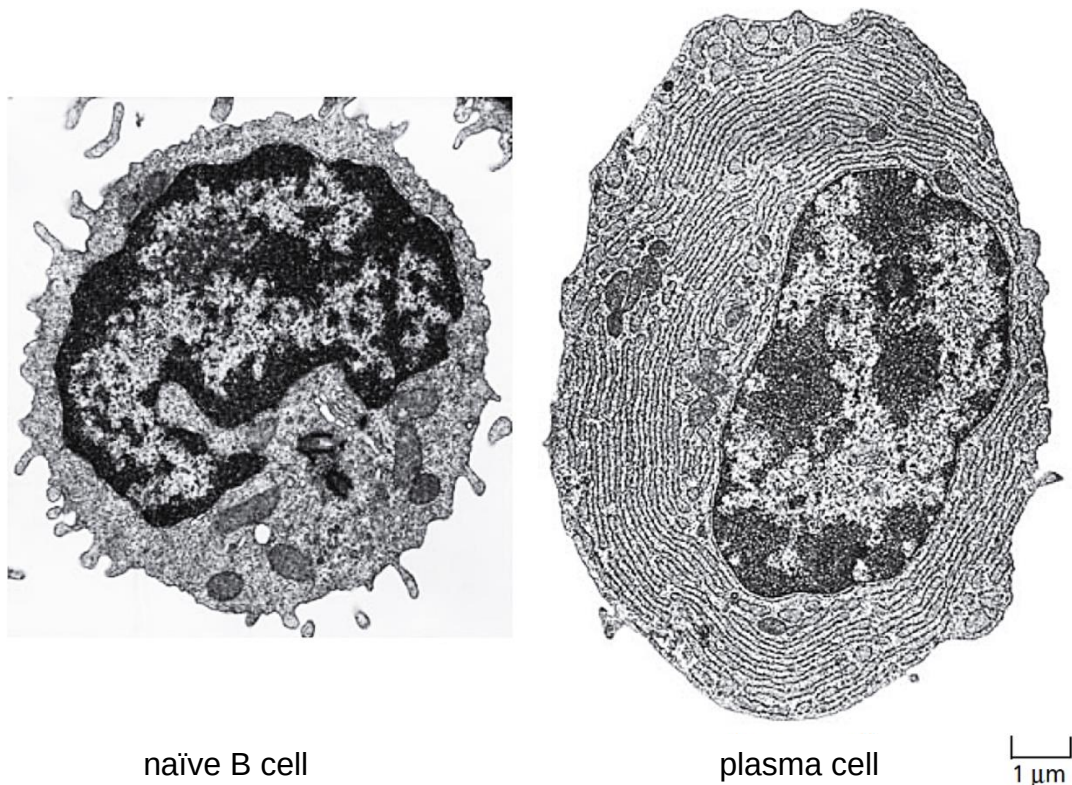


Fig. 10.1

- (c) Apart from the shape and size of the cells, state one other structural difference between a naïve B cell and a plasma cell **and** explain why this difference exists. [2]

1. Large number of rough endoplasmic reticulum in plasma cells.
2. Plasma cells synthesize large amount of antibodies (proteins), hence need more RER for antibody synthesis

[Total: 11]